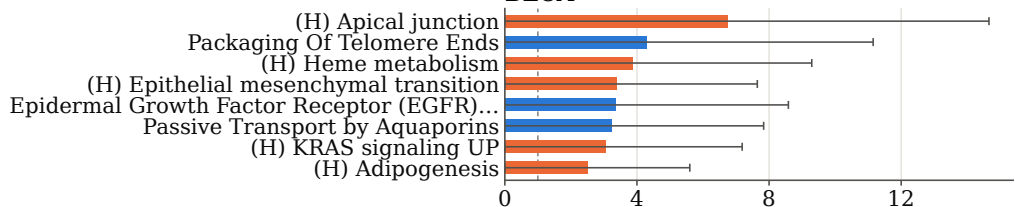
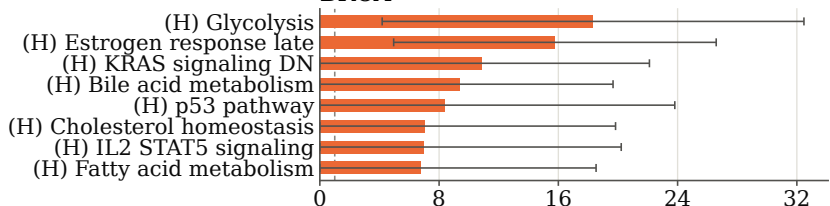


■ Reactome pathway ■ MSigDB Hallmark gene set (H)

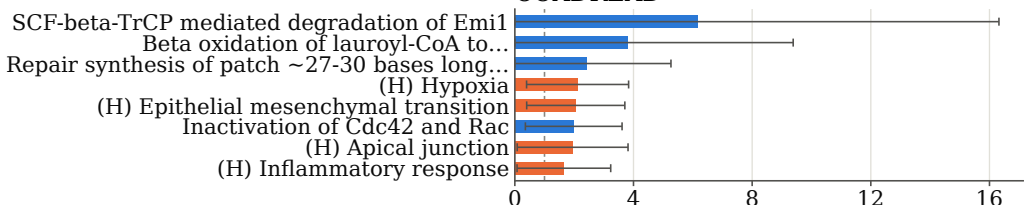
BLCA



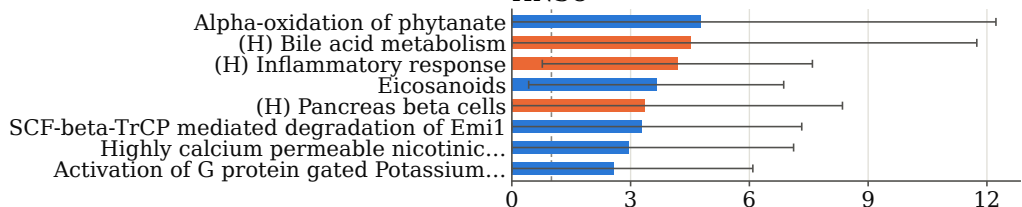
BRCA



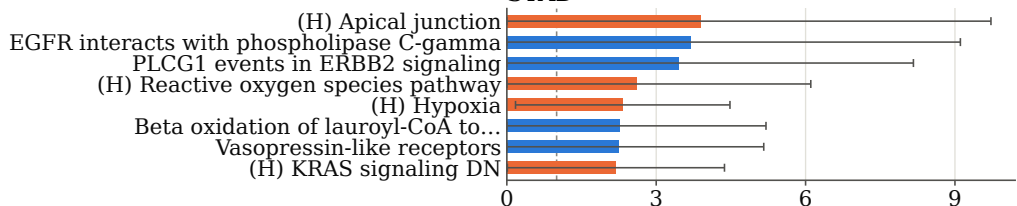
COADREAD



HNSC



STAD



Mean genomic cross-attention per pathway token ($\times$ uniform = 1/275)